

Product	Name	Reg. no.	Category
	ND-5000 Test Microprograms	211124B	STPR
Reason	☐ New product ☒ Error Correction ☒ Change/Addition ☐ Different Environment		
Documen- tation	Title	Reg. no.	
	SEMICS User Guide	20.024.2	
	TPE Monitor User Guide	30.105.1	
Purpose	Micro coded test programs for ND-5000 series CPU .		
Prerequi- sites	Computer	Type	Floating format Op. system Version
	ND-5000	All	All SIN III VSX/500 K Work mode 406
	Minimum mass storage resources for installation User User space Number of files ND5000-SEMICS 860 pages on 64 files		
Prerequi- sites	Minimum permanent mass storage resources User User space Number of files ND5000-SEMICS 860 pages on 64 files		
Product consists of			Reg. no. Source 250245B

Product	Name	Reg. no.	Category
	ND-5000 Test Microprograms	211124B	STPR

File name	Type	Containing
TPE-MON-100-B<rev>	PROG	TPE monitor
SEMICS-1-B<rev>	TEST	Main command overlay
SEMICS-2-B<rev>	NEXT	Second command overlay
SEMICS-3-B<rev>	NEXT	Third command overlay
SEMICS-ABBRV-B<rev>	DATA	Abbreviation list
INST-SEMICS-B<rev>	MODE	Copy TPE + SEMICS to disk
INST-EXT-B<rev>	MODE	Copy ext. error files to disk
ALU-VERIFY-B<rev>	EXT	Extended error information files
MIC-REGISTER-B<rev>	EXT	
MIC-SEQUENCE-B<rev>	EXT	
ALU-CARD-B<rev>	EXT	
CACHE-TEST-B<rev>	EXT	
IAC-TEST-B<rev>	EXT	
DAC-TEST-B<rev>	EXT	
IDU-REGISTER-B<rev>	EXT	
IDA-VERIFY-B<rev>	EXT	
DMM-REGISTER-B<rev>	EXT	
IMM-REGISTER-B<rev>	EXT	
MM-VERIFY-B<rev>	EXT	
AAP1-2-TEST-B<rev>	EXT	
AAP2-VERIFY-B<rev>	EXT	
INST-TESTS-B<rev>	MODE	Copy micro tests to disk
ALU-VERIFY-B<rev>	TEST	Files containing test microprograms
ALU-VERIFY-B<rev>	LABE	
ALU-VERIFY-B<rev>	DATA	
MIC-REGISTER-B<rev>	TEST	
MIC-REGISTER-B<rev>	LABE	
MIC-REGISTER-B<rev>	DATA	
MIC-SEQUENCE-B<rev>	TEST	
MIC-SEQUENCE-B<rev>	LABE	
MIC-SEQUENCE-B<rev>	DATA	
ALU-CARD-B<rev>	TEST	
ALU-CARD-B<rev>	LABE	
ALU-CARD-B<rev>	DATA	
CACHE-TEST-B<rev>	TEST	
CACHE-TEST-B<rev>	LABE	
CACHE-TEST-B<rev>	DATA	
IAC-TEST-B<rev>	TEST	
IAC-TEST-B<rev>	LABE	
IAC-TEST-B<rev>	DATA	
DAC-TEST-B<rev>	TEST	
DAC-TEST-B<rev>	LABE	
DAC-TEST-B<rev>	DATA	
BOOSTER-TEST-B<rev>	TEST	
BOOSTER-TEST-B<rev>	LABE	
BOOSTER-TEST-B<rev>	DATA	
IDU-REGISTER-B<rev>	TEST	
IDU-REGISTER-B<rev>	LABE	
IDU-REGISTER-B<rev>	DATA	
IDA-VERIFY-B<rev>	TEST	
IDA-VERIFY-B<rev>	LABE	
IDA-VERIFY-B<rev>	DATA	

Product	Name	Reg. no.	Category
	ND-5000 Test Microprograms	211124B	STPR

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DMM-REGISTER-B<rev> TEST
DMM-REGISTER-B<rev> LABE
DMM-REGISTER-B<rev> DATA
IMM-REGISTER-B<rev> TEST
IMM-REGISTER-B<rev> LABE
IMM-REGISTER-B<rev> DATA
MM-VERIFY-B<rev>      TEST
MM-VERIFY-B<rev>      LABE
MM-VERIFY-B<rev>      DATA
AAP1-2-TEST-B<rev>    TEST
AAP1-2-TEST-B<rev>    LABE
AAP1-2-TEST-B<rev>    DATA
AAP2-VERIFY-B<rev>    TEST
AAP2-VERIFY-B<rev>    LABE
AAP2-VERIFY-B<rev>    DATA

```

1 NOTE:

<rev> is to be replaced by the current revision of the DIRECTORY or FILE. Revision is found on the "ND SOFTWARE LIBRARY DISKETTE" pages.

2 ERRORS CORRECTED

- LOOK-AT-REGISTER: It was impossible to write a value into X4 register.
- Prior to starting each test, the AIB register is checked for valid content. If it contains valid data, it is cleared to inhibit false data for the test.
- MM verify, tests 5 and 6 wrote in MF memory outside reserved area.
- On CPUs of type ND-5800, the following tests failed in the previous version of SEMICS: Cache test, tests 7 and 16-20; IDA verify, test 1; IDU register, tests 15, 17, 20 and 22; DAC test, test 26. These tests are now corrected.
- In some special cases (PAL specs., Temperature etc.), Cache test, tests 1 and 8-12 may fail in version A00 of SEMICS. This is now corrected.

Date 88.04.12		Norsk Data A.S PROGRAM DESCRIPTION		Page 4 of 9
Product	Name	Reg. no.	Category	
	ND-5000 Test Microprograms	211124B	STPR	

3 KNOWN, BUT NOT CORRECTED ERRORS

- The LOOK-AT-CACHE command: On CPUs of type ND-5800, the fields 'LA' and 'DDIR' in data cache are not valid.

4 CHANGED PREREQUISITES

- This version needs work mode 406, or newer, of SINTRAN for ND-5000.
- The ACCP PROMS should be dated 06.07.87 or later.

5 MODIFICATIONS

5.1 CHANGED COMMANDS

- The LOOK-AT-CONTROL-STORE and LOOK-AT-MEMORY commands now display next location after a deposit. In version A00 of SEMICS, the current location was redisplayed.
- The LOOK-AT-CONTROL-STOREs subcommands which access files are changed. The files will now be searched under current user (SYSTEM) if no user name is specified.
- The LOOK-AT-CONTROL-STOREs disassembler is now able to disassemble the mnemonic 'NEXT*'.
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5.2 NEW COMMANDS

- CHANGE-CPU is used to move to another CPU in multi-CPU systems (ND-5900). This command runs only from SINTRAN work mode 500.
- EXPLAIN-ABBREVIATION is implemented. It explains words which are used in the ND-5000 terminology.
- CONTROL-STORE-TEST is implemented.
- NEWS command is implemented, telling about changes from previous versions.

Date 88.04.12		Norsk Data A.S PROGRAM DESCRIPTION		Page 5 of 9	
Product	Name	Reg. no.	Category		
	ND-5000 Test Microprograms	211124B	STPR		

- EXECUTE-MACRO-INSTRUCTION is implemented. It makes it possible to execute ND-5000 instructions from SEMICS. The user must first find the octal code for the instructions, load the necessary registers etc.

5.3 IMPROVED PERFORMANCE

- The IDU register, DMM register and IMM register test systems contain new input data tables. The execution times are shorter.
- The IDAC (Booster) test is implemented.
- The QUICK mode parameter is implemented. It can be used for all tests, including the control store test to perform a 'fast run' of the tests.

Date 88.04.12		Norsk Data A.S PROGRAM DESCRIPTION	Page 6 of 9
Product	Name	Reg. no.	Category
	ND-5000 Test Microprograms	211124B	STPR

6 INSTALLATION PROCEDURE

ND-5000 Test Microprograms is delivered on two double sided/double density floppy diskettes.

- 1) The floppy diskette 211124B<rev>-XX-01D contains :
 - TPE monitor
 - SEMICS command overlays
 - Abbreviation table
 - Extended Error Information
- 2) The floppy diskette 211124B<rev>-XX-02D contains the collection of microtests.

The microtests can be run in two ways:

- 1) From the hard disk.
- 2) From the floppy disk.

We recommend you to copy all files to the hard disk. The other way, which is slower, should only be used when you are out of disk space. In both cases the TPE monitor and SEMICS itself must be loaded to the hard disk. If the extended error information is to be used, the microtests must be run from the hard disk too. (Regarding extended error information, see the SEMICS User Guide (20.024.2).)

6.1 Preparations for running SEMICS.

- 1) Running SEMICS from the hard disk requires approximately 155 pages. (TPE monitor, SEMICS command overlays and the abbreviation table.)
- 2) Running microtests from the hard disk requires approximately:
 - 710 pages if the extended error information is not used.
 - 855 pages if the extended error information is used.

6.1.1 Loading the TPE monitor and SEMICS command overlays.

- 1) Log in as user SYSTEM.
- 2) Create the user ND5000-SEMICS.

@CREATE-USER ND5000-SEMICS↓

Date 88.04.12		Norsk Data A.S PROGRAM DESCRIPTION	Page 7 of 9
Product	Name	Reg. no.	Category
	ND-5000 Test Microprograms	211124B	STPR

- 3) Give the number of pages required for the way you want to run the microtests (155, 710 or 855).

@GIVE-USER-SPACE ND5000-SEMICS,<number of pages>↓

- 4) Log in as user ND5000-SEMICS.

- 5) Insert diskette 211124B<rev>-XX-01D (with TPE, SEMICS and Extended Error Information) and enter the directory with the command:

@ENTER-DIRECTORY 211124B<rev>-XX-01D,<device name>,<device unit>↓

- 6) Install SEMICS by running the mode file:

@MODE (211124B<rev>-XX-01D:FLOPPY-USER)INST-SEMICS-B<rev>:-MODE,,,↓

This mode job will copy the SEMICS command overlays, the abbreviation table and the TPE Monitor to user ND5000-SEMICS on the hard disk.

- 7) If you are going to use the Extended Error Information, install the files by running the mode file:

@MODE (211124B<rev>-XX-01D:FLOPPY-USER)INST-EXT-B<rev>:-MODE,,,↓

This mode job will copy all :EXT files to user ND5000-SEMICS on the hard disk.

- 8) Release the directory with the command:

@RELEASE-DIRECTORY 211124B<rev>-XX-01D↓

6.1.2 Copying the test microprograms to the hard disk.

This section describes how to install the microtests on the hard disk. You will not need the floppy diskettes after this installation.

- 1) Log in as user ND5000-SEMICS.
- 2) Ensure that the user ND5000-SEMICS has enough pages.
- 3) Insert diskette 211124B<rev>-XX-02D (with the microtests) and enter the directory with the command:

@ENTER-DIRECTORY 211124B<rev>-XX-02D,<device name>,<device unit>↓

Date 88.04.12		Norsk Data A.S PROGRAM DESCRIPTION	Page 8 of 9
Product	Name	Reg. no.	Category
	ND-5000 Test Microprograms	211124B	STPR

- 4) Install the microtests by running the mode file:

```
@MODE (211124B<rev>-XX-02D:FLOPPY-USER)INST-TESTS-B<rev>:-  
MODE,,,↓
```

This mode job will copy all testprogram files to user ND5000-SEMICS on the hard disk.

- 5) Release the directory with the command:

```
@RELEASE-DIRECTORY 211124B<rev>-XX-02D↓
```

You can now start the TPE monitor and then run SEMICS. Instructions on how to start the TPE monitor are given in the section "Starting the TPE monitor". After starting the TPE monitor you can load and run SEMICS. This is described in the SEMICS User Guide (Reg.no. 20.024.2).

6.1.3 Prepare running the test microprograms from the floppy diskette.

As long as the TPE monitor and SEMICS are loaded in your system, you can run microtests directly from the floppy diskette.

- 1) Log in as user SYSTEM.
- 2) Insert diskette 211124B<rev>-XX-02D (with microtests) and enter the directory with the command:

```
@ENTER-DIRECTORY 211124B<rev>-XX-02D,<device name>,<device  
unit>↓
```

You can now start the TPE monitor and then run SEMICS. Instructions on how to start TPE monitor are given in the section "Starting the TPE monitor". After starting the TPE monitor you can load and run SEMICS.

NOTE: The Extended Error Information cannot be used when running microtests from the floppy diskette.

Date 88.04.12		Norsk Data A.S PROGRAM DESCRIPTION		Page 9 of 9
Product	Name	Reg. no.	Category	
	ND-5000 Test Microprograms	211124B	STPR	

6.2 Starting the TPE monitor.

To start the TPE monitor:

- 1) Log in as user SYSTEM.
- 2) Type the following:

@(ND5000-SEMICS)TPE-MON-100-:PROG↓

When the TPE monitor has started, it will identify itself with this heading:

TPE Monitor, ND-100 / ND-110 - Version: lnn - yyyy-mm-dd

The TPE monitor is now ready to accept your commands.

6.3 Starting SEMICS.

When the TPE monitor is running, you can load the test program as follows:

>Load (ND5000-SEMICS)SEMICS↓

The program will identify itself and the SEMICS program commands will be available.

Remember that the first command used after starting SEMICS is INVESTIGATE-AND-RESERVE.

Exeption: When the ND-5000 CPU has very poor HW status it may be impossible to run this command. System parameters are then set with the command SET-PARAMETER.

If you are running the microtests from the floppy diskette, after entering the floppy diskette 211124B<rev>-XX-02D, you must specify directory and user name when you run the first test. Example:

TPE>RUN↓

Test system: (211124B<rev>-XX-02D:FLOPPY-USER)<Test name>↓

The commands NEWS and INFO may be useful. They give information about the current version of SEMICS.

ND SOFTWARE LIBRARY DISKETTE

PAGE 1

Containing :
ND-5000 Test Microprograms

Directory Name :
211124B00-XX-01D

User Name : FLOPPY-USER

No	File name	Type	T	Public	Friend	Own	Pages	Bytes
0	TPE-MON-100-B00	PROG	I	RWACD	RWACD	RWACD	29	56320
1	SEMICs-1-B00	TEST	I	RWACD	RWACD	RWACD	39	76288
2	SEMICs-2-B00	NEXT	I	RWACD	RWACD	RWACD	28	54272
3	SEMICs-3-B00	NEXT	I	RWACD	RWACD	RWACD	28	54272
4	INST-EXT-B00	MODE	I	RWACD	RWACD	RWACD	2	1669
5	INST-SEMICs-B00	MODE	I	RWACD	RWACD	RWACD	4	627
6	SEMICs-ABBRV-B00	DATA	I	RWACD	RWACD	RWACD	26	50641
7	ALU-VERIFY-B00	EXT	I	R	RWACD	RWACD	4	6144
8	MIC-REGISTER-B00	EXT	I	R	RWACD	RWACD	5	8192
9	MIC-SEQUENCE-B00	EXT	I	R	RWACD	RWACD	2	2048
10	ALU-CARD-B00	EXT	I	R	RWACD	RWACD	22	43008
11	CACHE-TEST-B00	EXT	I	R	RWACD	RWACD	39	77824
12	IAC-TEST-B00	EXT	I	R	RWACD	RWACD	5	8192
13	DAC-TEST-B00	EXT	I	R	RWACD	RWACD	9	16384
14	IDU-REGISTER-B00	EXT	I	R	RWACD	RWACD	6	10240
15	IDA-VERIFY-B00	EXT	I	R	RWACD	RWACD	5	8192
16	DMM-REGISTER-B00	EXT	I	R	RWACD	RWACD	10	18432
17	IMM-REGISTER-B00	EXT	I	R	RWACD	RWACD	10	18432
18	MM-VERIFY-B00	EXT	I	R	RWACD	RWACD	8	14336
19	AAP1-2-TEST-B00	EXT	I	R	RWACD	RWACD	13	24576
20	AAP2-VERIFY-B00	EXT	I	R	RWACD	RWACD	4	6144
21	files using 298 pages. 610 pages reserved out of 610 pages.							

ND SOFTWARE LIBRARY DISKETTE

PAGE 1

Containing :
ND-5000 Test Microprograms

Directory Name :
211124B00-XX-02D

User Name : FLOPPY-USER

No	File name	Type	T	Public	Friend	Own	Pages	Bytes
0	INST-TESTS-B00	MODE	I	RWACD	RWACD	RWACD	4	5416
1	ALU-VERIFY-B00	DATA	I	R	RWACD	RWACD	13	22880
2	ALU-VERIFY-B00	LABE	I	R	RWACD	RWACD	2	1608
3	MIC-REGISTER-B00	DATA	I	R	RWACD	RWACD	6	8544
4	MIC-REGISTER-B00	LABE	I	R	RWACD	RWACD	2	744
5	MIC-SEQUENCE-B00	DATA	I	R	RWACD	RWACD	5	7296
6	MIC-SEQUENCE-B00	LABE	I	R	RWACD	RWACD	2	552
7	ALU-CARD-B00	DATA	I	R	RWACD	RWACD	11	19392
8	ALU-CARD-B00	LABE	I	R	RWACD	RWACD	2	960
9	CACHE-TEST-B00	DATA	I	R	RWACD	RWACD	52	102496
10	CACHE-TEST-B00	LABE	I	R	RWACD	RWACD	9	16200
11	IAC-TEST-B00	DATA	I	R	RWACD	RWACD	7	10688
12	IAC-TEST-B00	LABE	I	R	RWACD	RWACD	2	456
13	DAC-TEST-B00	DATA	I	R	RWACD	RWACD	11	19504
14	DAC-TEST-B00	LABE	I	R	RWACD	RWACD	2	936
15	IDU-REGISTER-B00	DATA	I	R	RWACD	RWACD	10	18048
16	IDU-REGISTER-B00	LABE	I	R	RWACD	RWACD	2	1440
17	IDA-VERIFY-B00	DATA	I	R	RWACD	RWACD	26	50640
18	IDA-VERIFY-B00	LABE	I	R	RWACD	RWACD	6	8400
19	BOOSTER-TEST-B00	DATA	I	R	RWACD	RWACD	5	7040
20	BOOSTER-TEST-B00	LABE	I	R	RWACD	RWACD	2	984
21	DMM-REGISTER-B00	DATA	I	R	RWACD	RWACD	7	12256
22	DMM-REGISTER-B00	LABE	I	R	RWACD	RWACD	2	792
23	IMM-REGISTER-B00	DATA	I	R	RWACD	RWACD	7	10544
24	IMM-REGISTER-B00	LABE	I	R	RWACD	RWACD	2	792
25	MM-VERIFY-B00	DATA	I	R	RWACD	RWACD	22	42400
26	MM-VERIFY-B00	LABE	I	R	RWACD	RWACD	4	5352
27	AAP1-2-TEST-B00	DATA	I	R	RWACD	RWACD	11	20128
28	AAP1-2-TEST-B00	LABE	I	R	RWACD	RWACD	2	1440
29	AAP2-VERIFY-B00	DATA	I	R	RWACD	RWACD	10	17120

CONTINUED

ND SOFTWARE LIBRARY DISKETTE

PAGE 2

Containing :
ND-5000 Test Microprograms

Directory Name :
211124B00-XX-02D

User Name : FLOPPY-USER

No	File name	Type	T	Public	Friend	Own	Pages	Bytes
30	AAP2-VERIFY-B00	LABE	I	R	RWACD	RWACD	3	3744
31	ALU-VERIFY-B00	TEST	I	R	RWACD	RWACD	5	135168
32	MIC-REGISTER-B00	TEST	I	R	RWACD	RWACD	9	139264
33	MIC-SEQUENCE-B00	TEST	I	R	RWACD	RWACD	8	139264
34	ALU-CARD-B00	TEST	I	R	RWACD	RWACD	32	180224
35	CACHE-TEST-B00	TEST	I	R	RWACD	RWACD	24	165888
36	IAC-TEST-B00	TEST	I	R	RWACD	RWACD	14	143360
37	DAC-TEST-B00	TEST	I	R	RWACD	RWACD	45	200704
38	IDU-REGISTER-B00	TEST	I	R	RWACD	RWACD	34	182272
39	IDA-VERIFY-B00	TEST	I	R	RWACD	RWACD	4	133120
40	BOOSTER-TEST-B00	TEST	I	R	RWACD	RWACD	6	10240
41	DMM-REGISTER-B00	TEST	I	R	RWACD	RWACD	17	147456
42	IMM-REGISTER-B00	TEST	I	R	RWACD	RWACD	17	147456
43	MM-VERIFY-B00	TEST	I	R	RWACD	RWACD	19	157696
44	AAP1-2-TEST-B00	TEST	I	R	RWACD	RWACD	67	237568
45	AAP2-VERIFY-B00	TEST	I	R	RWACD	RWACD	9	141312
46	files using 561 pages. 610 pages reserved out of 610 pages.							